

CO4-AT1-CSA0925 - JAVA PROGRAMMING

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QUESTION 1: EMPLOYEE PAYROLLDASHBOARD

Code:

```
import java.awt.*;
import java.awt.event.*;
import java.util.ArrayList;
public class EmployeePayrollDashboard extends Frame
implements ActionListener {

    private TextField tfEmpId, tfEmpName, tfBasicSalary,
tfWorkingDays;

    private Choice choiceDept, choiceDesignation;
    private CheckboxGroup ratingGroup
```

```
private Checkbox cbExcellent, cbGood, cbAverage,  
cbPoor;
```

```
private Checkbox cbHRA, cbDA, cbMedical;
```

```
// ---- Output / action components ----
```

```
private List employeeList;
```

```
private TextArea reportArea;
```

```
private Button btnGenerate, btnReset, btnExit;
```

```
private ArrayList<String> employeeRecords = new  
ArrayList<>();
```

```
public EmployeePayrollDashboard() {  
    super("Employee Performance & Payroll Dashboard");  
    setLayout(new BorderLayout(10, 10));  
  
    add(buildFormPanel(), BorderLayout.NORTH);  
    add(buildListPanel(), BorderLayout.WEST);
```

```
add(buildReportPanel(), BorderLayout.CENTER);  
add(buildButtonPanel(), BorderLayout.SOUTH);
```

```
// Allow closing the window via the OS close button
```

```
addWindowListener(new WindowAdapter() {  
    public void windowClosing(WindowEvent e) {  
        dispose();  
        System.exit(0);  
    }  
});
```

```
setSize(760, 620);  
setLocationRelativeTo(null);  
setVisible(true);  
}
```

```
private Panel buildFormPanel() {  
    Panel formPanel = new Panel(new GridBagLayout());  
    formPanel.setBackground(new Color(235, 242, 250));  
}
```

```
GridBagConstraints gbc = new GridBagConstraints();  
gbc.insets = new Insets(6, 8, 6, 8);  
gbc.anchor = GridBagConstraints.WEST;  
gbc.fill = GridBagConstraints.HORIZONTAL;
```

```
int row = 0;
```

```
gbc.gridx = 0; gbc.gridy = row;  
formPanel.add(new Label("Employee ID:"), gbc);  
tfEmpId = new TextField(15);  
gbc.gridx = 1; formPanel.add(tfEmpId, gbc);
```

```
gbc.gridx = 2;  
formPanel.add(new Label("Employee Name:"), gbc);  
tfEmpName = new TextField(15);  
gbc.gridx = 3; formPanel.add(tfEmpName, gbc);  
row++;
```

```
// Department (Choice)
```

```
gbc.gridx = 0; gbc.gridy = row;  
formPanel.add(new Label("Department:"), gbc);
```

```
choiceDept = new Choice();
choiceDept.add("IT");
choiceDept.add("HR");
choiceDept.add("Finance");
choiceDept.add("Operations");
choiceDept.add("Sales");
gbc.gridx = 1; formPanel.add(choiceDept, gbc);

// Designation (Choice)
gbc.gridx = 2;
formPanel.add(new Label("Designation:"), gbc);
choiceDesignation = new Choice();
choiceDesignation.add("Software Engineer");
choiceDesignation.add("Senior Engineer");
choiceDesignation.add("Team Lead");
choiceDesignation.add("Manager");
choiceDesignation.add("Intern");
gbc.gridx = 3; formPanel.add(choiceDesignation, gbc);
row++;

// Basic Salary
gbc.gridx = 0; gbc.gridy = row;
formPanel.add(new Label("Basic Salary (Rs.):"), gbc);
```

```
tfBasicSalary = new TextField(15);
gbc.gridx = 1; formPanel.add(tfBasicSalary, gbc);

// Working Days
gbc.gridx = 2;
formPanel.add(new Label("Working Days:"), gbc);
tfWorkingDays = new TextField(15);
gbc.gridx = 3; formPanel.add(tfWorkingDays, gbc);
row++;
```

```
// Performance Rating (CheckboxGroup = radio buttons)
gbc.gridx = 0; gbc.gridy = row;
formPanel.add(new Label("Performance Rating:"), gbc);
```

```
Panel ratingPanel = new Panel(new
FlowLayout(FlowLayout.LEFT, 8, 0));
ratingGroup = new CheckboxGroup();
cbExcellent = new Checkbox("Excellent", ratingGroup,
true);
cbGood = new Checkbox("Good", ratingGroup, false);
cbAverage = new Checkbox("Average", ratingGroup,
false);
cbPoor = new Checkbox("Poor", ratingGroup, false);
ratingPanel.add(cbExcellent);
```

```
ratingPanel.add(cbGood);  
ratingPanel.add(cbAverage);  
ratingPanel.add(cbPoor);
```

```
gbc.gridx = 1; gbc.gridwidth = 3;  
formPanel.add(ratingPanel, gbc);  
gbc.gridwidth = 1;  
row++;
```

```
// Allowances (independent Checkboxes)  
gbc.gridx = 0; gbc.gridy = row;  
formPanel.add(new Label("Allowances:"), gbc);
```

```
Panel allowancePanel = new Panel(new  
FlowLayout(FlowLayout.LEFT, 8, 0));  
cbHRA = new Checkbox("HRA (20%)", true);  
cbDA = new Checkbox("DA (10%)", true);  
cbMedical = new Checkbox("Medical (Rs.1000)", true);  
allowancePanel.add(cbHRA);  
allowancePanel.add(cbDA);  
allowancePanel.add(cbMedical);  
  
gbc.gridx = 1; gbc.gridwidth = 3;
```

```
formPanel.add(allowancePanel, gbc);  
gbc.gridwidth = 1;  
  
return formPanel;  
}
```

```
private Panel buildListPanel() {  
    Panel panel = new Panel(new BorderLayout(5, 5));  
    panel.add(new Label("Employee Records:"),  
BorderLayout.NORTH);  
    employeeList = new List(10, false);  
    panel.add(employeeList, BorderLayout.CENTER);  
    panel.setPreferredSize(new Dimension(220, 200));  
    return panel;  
}
```

```
private Panel buildReportPanel() {  
    Panel panel = new Panel(new BorderLayout(5, 5));  
    panel.add(new Label("Payroll Report:"),  
BorderLayout.NORTH);  
    reportArea = new TextArea(15, 50);  
    reportArea.setEditable(false);  
    reportArea.setFont(new Font("Monospaced",  
Font.PLAIN, 12));  
}
```



```
    panel.add(reportArea, BorderLayout.CENTER);  
    return panel;  
}
```

```
private Panel buildButtonPanel() {  
    Panel panel = new Panel(new  
FlowLayout(FlowLayout.CENTER, 15, 10));  
    btnGenerate = new Button("Generate Report");  
    btnReset = new Button("Reset");  
    btnExit = new Button("Exit");  
  
    btnGenerate.addActionListener(this);  
    btnReset.addActionListener(this);  
    btnExit.addActionListener(this);  
  
    panel.add(btnGenerate);  
    panel.add(btnReset);  
    panel.add(btnExit);  
    return panel;  
}
```

```
// -----
```

```
// Event handling
```

```
// -----
```

```
@Override
```

```
public void actionPerformed(ActionEvent e) {
```

```
    Object src = e.getSource();
```

```
    if (src == btnGenerate) {
```

```
        generateReport();
```

```
    } else if (src == btnReset) {
```

```
        resetForm();
```

```
    } else if (src == btnExit) {
```

```
        dispose();
```

```
        System.exit(0);
```

```
    }
```

```
}
```

```
private void generateReport() {
```

```
    String empId = tfEmpId.getText().trim();
```

```
    String empName = tfEmpName.getText().trim();
```

```
    String basicText = tfBasicSalary.getText().trim();
```

```
    String daysText = tfWorkingDays.getText().trim();
```

```
// ---- Validation ----
```

```
        if (empId.isEmpty() || empName.isEmpty() ||
basicText.isEmpty() || daysText.isEmpty()) {
            showMessage("Validation Error: Please fill in all fields
(ID, Name, Salary, Working Days).");
            return;
        }
```

```
        double basicSalary;
        int workingDays;
        try {
            basicSalary = Double.parseDouble(basicText);
            if (basicSalary <= 0) {
                showMessage("Validation Error: Basic Salary must
be a positive number.");
                return;
            }
        } catch (NumberFormatException ex) {
            showMessage("Validation Error: Basic Salary must be
numeric.");
            return;
        }
```

```
        try {
            workingDays = Integer.parseInt(daysText);
```

```

        if (workingDays < 0 || workingDays > 31) {
            showMessage("Validation Error: Working Days
must be between 0 and 31.");
            return;
        }
    } catch (NumberFormatException ex) {
        showMessage("Validation Error: Working Days must
be a whole number.");
        return;
    }
}

```

```

String department = choiceDept.getSelectedItem();
String designation =
choiceDesignation.getSelectedItem();
String rating =
ratingGroup.getSelectedCheckbox().getLabel();

```

```

// ---- Business rule calculations ----
double proRatedBasic = (workingDays < 26)
    ? (basicSalary / 26.0) * workingDays
    : basicSalary;

double hra = cbHRA.getState() ? proRatedBasic * 0.20 :
0;

```

```
double da = cbDA.getState() ? proRatedBasic * 0.10 : 0;
double medical = cbMedical.getState() ? 1000 : 0;
double grossSalary = proRatedBasic + hra + da +
medical;
```

```
double bonusPercent;
switch (rating) {
    case "Excellent": bonusPercent = 0.15; break;
    case "Good":      bonusPercent = 0.10; break;
    case "Average":   bonusPercent = 0.05; break;
    default:          bonusPercent = 0.0;
}
double performanceBonus = proRatedBasic *
bonusPercent;
```

```
double pfDeduction = proRatedBasic * 0.12;
double netSalary = grossSalary + performanceBonus -
pfDeduction;
```

```
// ---- Build report entry ----
String summary = String.format("%s | %s | %s | Net:
Rs.%.2f", empId, empName, department, netSalary);
employeeRecords.add(summary);
employeeList.add(summary);
```

```

StringBuffer sb = new StringBuffer();

sb.append("=====
=====\\n");

sb.append("      EMPLOYEE PAYROLL REPORT\\n");

sb.append("=====
=====\\n");

sb.append(String.format("Employee ID      : %s\\n",
empId));

sb.append(String.format("Employee Name    : %s\\n",
empName));

sb.append(String.format("Department      : %s\\n",
department));

sb.append(String.format("Designation     : %s\\n",
designation));

sb.append(String.format("Working Days    : %d\\n",
workingDays));

sb.append(String.format("Performance Rating: %s\\n",
rating));

sb.append("-----\\n");

sb.append(String.format("Pro-rated Basic  :
Rs.%.2f\\n", proRatedBasic));

sb.append(String.format("HRA              : Rs.%.2f\\n",
hra));

sb.append(String.format("DA               : Rs.%.2f\\n",
da));

```

```

        sb.append(String.format("Medical Allowance :
Rs.%.2f%n", medical));

        sb.append(String.format("Gross Salary      : Rs.%.2f%n",
grossSalary));

        sb.append(String.format("Performance Bonus : Rs.%.2f
(%.0f%%)%n", performanceBonus, bonusPercent * 100));

        sb.append(String.format("PF Deduction (12%%):
Rs.%.2f%n", pfDeduction));

        sb.append("-----\n");

        sb.append(String.format("NET SALARY      :
Rs.%.2f%n", netSalary));

        sb.append("=====\n\n");

reportArea.append(sb.toString());
reportArea.setCaretPosition(reportArea.getText().length(
));
}

```

```

private void resetForm() {
    tfEmpId.setText("");
    tfEmpName.setText("");
    tfBasicSalary.setText("");
    tfWorkingDays.setText("");
    choiceDept.select(0);
}

```

```

        choiceDesignation.select(0);
        ratingGroup.setSelectedCheckbox(cbExcellent);
        cbHRA.setState(true);
        cbDA.setState(true);
        cbMedical.setState(true);
        tfEmpId.requestFocus();
        // Note: Reset only clears the input form.
        // employeeList / reportArea history is intentionally kept.
    }

    private void showMessage(String msg) {
        reportArea.append(">> " + msg + "\n\n");
        reportArea.setCaretPosition(reportArea.getText().length(
));
    }

    public static void main(String[] args) {
        EventQueue.invokeLater(EmployeePayrollDashboard::n
ew);
    }
}

```

OUTPUT:

Employee Performance & Payroll Dashboard

Employee ID: Employee Name:

Department: Designation:

Basic Salary (Rs.): Working Days:

Performance Rating: ☒ Excellent ☐ Good ☐ Average ☐ Poor

Allowances: ☒ HRA (20%) ☒ DA (10%) ☒ Medical (Rs.1000)

Employee Records: 1432 | Deepak | IT | Net: Rs.538115.38

Payroll Report:

```

=====
EMPLOYEE PAYROLL REPORT
=====
Employee ID      : 1432
Employee Name    : Deepak
Department       : IT
Designation      : Software Engineer
Working Days     : 21
Performance Rating: Excellent
=====
Pro-rated Basic  : Rs.403846.15
HRA              : Rs.80769.23
DA              : Rs.40384.62
Medical Allowance: Rs.1000.00
Gross Salary     : Rs.526000.00
Performance Bonus: Rs.60576.92 (15%)
PF Deduction (12%): Rs.48461.54
=====
NET SALARY      : Rs.538115.38
=====

```

Generate Report Reset Exit

QUESTION 2 – ONLINE FOOD ORDER AND BILL GENRATING SYSTEM

CODE:

```

import java.awt.*;
import java.awt.event.*;
import java.util.HashMap;
import java.util.Map;

public class FoodOrderingBillingSystem extends Frame implements ActionListener {

    // ---- CardLayout management ----

    private CardLayout cardLayout;

    private Panel cardPanel;

```

```

private static final String CARD_CUSTOMER = "Customer/Order Details";
private static final String CARD_FOOD = "Food Selection";
private static final String CARD_BILL = "Bill/Order Summary";

// ---- Card 1: Customer/Order Details ----
private TextField tfCustomerName, tfTableNumber;

// ---- Card 2: Food Selection ----
private Choice choiceCategory, choiceFoodItem;
private TextField tfQuantity;
private List orderList;
private Button btnAddItem;

// ---- Card 3: Bill/Order Summary ----
private TextArea billArea;
private Button btnGenerateBill, btnClear;

// ---- Navigation buttons (shared) ----
private Button btnNext1, btnNext2, btnPrevious2, btnPrevious3;

// ---- Menu data: category -> {item -> price} ----
private Map<String, Map<String, Double>> menu = new HashMap<>();

// ---- Order data ----
private double subtotal = 0.0;
private static final double GST_RATE = 0.05;

public FoodOrderingBillingSystem() {
    super("Online Food Ordering and Billing System");
    buildMenuData();
}

```

```

cardLayout = new CardLayout();
cardPanel = new Panel(cardLayout);

cardPanel.add(buildCustomerCard(), CARD_CUSTOMER);
cardPanel.add(buildFoodSelectionCard(), CARD_FOOD);
cardPanel.add(buildBillCard(), CARD_BILL);

setLayout(new BorderLayout());
add(cardPanel, BorderLayout.CENTER);

addWindowListener(new WindowAdapter() {
    public void windowClosing(WindowEvent e) {
        dispose();
        System.exit(0);
    }
});

setSize(700, 520);
setLocationRelativeTo(null);
setVisible(true);
}

// -----
// Menu data setup
// -----

private void buildMenuData() {
    Map<String, Double> starters = new HashMap<>();
    starters.put("Spring Roll", 120.0);

```

```
starters.put("Paneer Tikka", 180.0);
starters.put("Chicken Wings", 220.0);
menu.put("Starters", starters);
```

```
Map<String, Double> mainCourse = new HashMap<>();
mainCourse.put("Veg Biryani", 180.0);
mainCourse.put("Chicken Biryani", 250.0);
mainCourse.put("Butter Naan", 40.0);
mainCourse.put("Paneer Butter Masala", 220.0);
menu.put("Main Course", mainCourse);
```

```
Map<String, Double> beverages = new HashMap<>();
beverages.put("Cold Coffee", 90.0);
beverages.put("Fresh Lime Soda", 60.0);
beverages.put("Mango Lassi", 80.0);
menu.put("Beverages", beverages);
```

```
Map<String, Double> desserts = new HashMap<>();
desserts.put("Gulab Jamun", 70.0);
desserts.put("Ice Cream", 90.0);
desserts.put("Brownie", 110.0);
menu.put("Desserts", desserts);
```

```
}
```

```
// -----
```

```
// Card 1: Customer / Order Details
```

```
// -----
```

```
private Panel buildCustomerCard() {
    Panel outer = new Panel(new BorderLayout());
```

```
Panel form = new Panel(new GridBagLayout());
GridBagConstraints gbc = new GridBagConstraints();
gbc.insets = new Insets(10, 10, 10, 10);
gbc.anchor = GridBagConstraints.WEST;
gbc.fill = GridBagConstraints.HORIZONTAL;

Label heading = new Label("STEP 1: Customer / Order Details", Label.CENTER);
heading.setFont(new Font("SansSerif", Font.BOLD, 16));
gbc.gridx = 0; gbc.gridy = 0; gbc.gridwidth = 2;
form.add(heading, gbc);
gbc.gridwidth = 1;

gbc.gridx = 0; gbc.gridy = 1;
form.add(new Label("Customer Name:"), gbc);
tfCustomerName = new TextField(20);
gbc.gridx = 1;
form.add(tfCustomerName, gbc);

gbc.gridx = 0; gbc.gridy = 2;
form.add(new Label("Table Number:"), gbc);
tfTableNumber = new TextField(20);
gbc.gridx = 1;
form.add(tfTableNumber, gbc);

outer.add(form, BorderLayout.CENTER);

Panel navPanel = new Panel(new FlowLayout(FlowLayout.RIGHT));
btnNext1 = new Button("Next >>");
btnNext1.addActionListener(this);
```

```

navPanel.add(btnNext1);
outer.add(navPanel, BorderLayout.SOUTH);

return outer;
}

// -----
// Card 2: Food Selection
// -----

private Panel buildFoodSelectionCard() {
    Panel outer = new Panel(new BorderLayout(5, 5));

    Panel form = new Panel(new GridBagLayout());
    GridBagConstraints gbc = new GridBagConstraints();
    gbc.insets = new Insets(8, 8, 8, 8);
    gbc.anchor = GridBagConstraints.WEST;
    gbc.fill = GridBagConstraints.HORIZONTAL;

    Label heading = new Label("STEP 2: Food Selection", Label.CENTER);
    heading.setFont(new Font("SansSerif", Font.BOLD, 16));
    gbc.gridx = 0; gbc.gridy = 0; gbc.gridwidth = 2;
    form.add(heading, gbc);
    gbc.gridwidth = 1;

    gbc.gridx = 0; gbc.gridy = 1;
    form.add(new Label("Food Category:"), gbc);
    choiceCategory = new Choice();
    for (String cat : menu.keySet()) {
        choiceCategory.add(cat);
    }
}

```

```

}

choiceCategory.addItemListener(e -> populateFoodItems());

gbc.gridx = 1;
form.add(choiceCategory, gbc);


gbc.gridx = 0; gbc.gridy = 2;
form.add(new Label("Food Item:"), gbc);
choiceFoodItem = new Choice();
gbc.gridx = 1;
form.add(choiceFoodItem, gbc);


gbc.gridx = 0; gbc.gridy = 3;
form.add(new Label("Quantity:"), gbc);
tfQuantity = new TextField("1", 20);
gbc.gridx = 1;
form.add(tfQuantity, gbc);


btnAddItem = new Button("Add Item");
btnAddItem.addActionListener(this);
gbc.gridx = 1; gbc.gridy = 4;
form.add(btnAddItem, gbc);


outer.add(form, BorderLayout.NORTH);


Panel listPanel = new Panel(new BorderLayout(5, 5));
listPanel.add(new Label("Current Order:"), BorderLayout.NORTH);
orderList = new List(8, false);
listPanel.add(orderList, BorderLayout.CENTER);
outer.add(listPanel, BorderLayout.CENTER);

```

```

Panel navPanel = new Panel(new FlowLayout(FlowLayout.RIGHT));
btnPrevious2 = new Button("<< Previous");
btnNext2 = new Button("Next >>");
btnPrevious2.addActionListener(this);
btnNext2.addActionListener(this);
navPanel.add(btnPrevious2);
navPanel.add(btnNext2);
outer.add(navPanel, BorderLayout.SOUTH);

// populate items for the first category by default
populateFoodItems();

return outer;
}

private void populateFoodItems() {
    choiceFoodItem.removeAll();
    String category = choiceCategory.getSelectedItem();
    Map<String, Double> items = menu.get(category);
    if (items != null) {
        for (String item : items.keySet()) {
            choiceFoodItem.add(item);
        }
    }
}

// -----
// Card 3: Bill / Order Summary
// -----

```



```
private Panel buildBillCard() {  
    Panel outer = new Panel(new BorderLayout(5, 5));  
  
    Label heading = new Label("STEP 3: Bill / Order Summary", Label.CENTER);  
    heading.setFont(new Font("SansSerif", Font.BOLD, 16));  
    outer.add(heading, BorderLayout.NORTH);  
  
    billArea = new TextArea(18, 55);  
    billArea.setEditable(false);  
    billArea.setFont(new Font("Monospaced", Font.PLAIN, 12));  
    outer.add(billArea, BorderLayout.CENTER);  
  
    Panel buttonPanel = new Panel(new FlowLayout(FlowLayout.CENTER, 15, 10));  
    btnPrevious3 = new Button("<< Previous");  
    btnGenerateBill = new Button("Generate Bill");  
    btnClear = new Button("Clear");  
  
    btnPrevious3.addActionListener(this);  
    btnGenerateBill.addActionListener(this);  
    btnClear.addActionListener(this);  
  
    buttonPanel.add(btnPrevious3);  
    buttonPanel.add(btnGenerateBill);  
    buttonPanel.add(btnClear);  
    outer.add(buttonPanel, BorderLayout.SOUTH);  
  
    return outer;  
}  
  
// -----
```

```
// Event handling
```

```
// -----
```

```
@Override
```

```
public void actionPerformed(ActionEvent e) {
```

```
    Object src = e.getSource();
```

```
    if (src == btnNext1) {
```

```
        if (tfCustomerName.getText().trim().isEmpty() ||  
tfTableNumber.getText().trim().isEmpty()) {
```

```
            showMessageDialog("Please enter Customer Name and Table Number before  
proceeding.");
```

```
            return;
```

```
        }
```

```
        cardLayout.show(cardPanel, CARD_FOOD);
```

```
    } else if (src == btnAddItem) {
```

```
        addItemToOrder();
```

```
    } else if (src == btnPrevious2) {
```

```
        cardLayout.show(cardPanel, CARD_CUSTOMER);
```

```
    } else if (src == btnNext2) {
```

```
        if (orderList.getItemCount() == 0) {
```

```
            showMessageDialog("Please add at least one item before generating the bill.");
```

```
            return;
```

```
        }
```

```
        cardLayout.show(cardPanel, CARD_BILL);
```

```
    } else if (src == btnPrevious3) {
```

```
        cardLayout.show(cardPanel, CARD_FOOD);
```

```
    } else if (src == btnGenerateBill) {  
        generateBill();  
  
    } else if (src == btnClear) {  
        clearOrder();  
    }  
}
```

```
private void addItemToOrder() {  
    String category = choiceCategory.getSelectedItem();  
    String item = choiceFoodItem.getSelectedItem();  
  
    if (item == null || item.isEmpty()) {  
        showMessageDialog("Please select a food item before adding.");  
        return;  
    }  
  
    String qtyText = tfQuantity.getText().trim();  
    int quantity;  
    try {  
        quantity = Integer.parseInt(qtyText);  
        if (quantity <= 0) {  
            showMessageDialog("Quantity must be a positive whole number.");  
            return;  
        }  
    } catch (NumberFormatException ex) {  
        showMessageDialog("Quantity must be a valid whole number.");  
        return;  
    }  
}
```

```

        double price = menu.get(category).get(item);
        double itemCost = price * quantity;
        subtotal += itemCost;

        String line = String.format("%s x%d @ Rs.%.2f = Rs.%.2f", item, quantity, price,
itemCost);
        orderList.add(line);

        tfQuantity.setText("1");
    }

    private void generateBill() {
        double gst = subtotal * GST_RATE;
        double finalAmount = subtotal + gst;

        StringBuilder sb = new StringBuilder();
        sb.append("=====\n");
        sb.append("        RESTAURANT FINAL BILL\n");
        sb.append("=====\n");
        sb.append(String.format("Customer Name : %s\n", tfCustomerName.getText().trim()));
        sb.append(String.format("Table Number : %s\n", tfTableNumber.getText().trim()));
        sb.append("-----\n");
        sb.append("Ordered Items:\n");
        String[] items = orderList.getItems();
        for (String item : items) {
            sb.append(" - ").append(item).append("\n");
        }
        sb.append("-----\n");
        sb.append(String.format("Subtotal      : Rs.%.2f\n", subtotal));
        sb.append(String.format("GST (5%%)      : Rs.%.2f\n", gst));
    }

```

```

sb.append("-----\n");
sb.append(String.format("FINAL AMOUNT : Rs.%.2f%n", finalAmount));
sb.append("=====\n");

billArea.setText(sb.toString());
}

```

```

private void clearOrder() {
    orderList.removeAll();
    billArea.setText("");
    subtotal = 0.0;
    tfCustomerName.setText("");
    tfTableNumber.setText("");
    tfQuantity.setText("1");
    cardLayout.show(cardPanel, CARD_CUSTOMER);
}

```

```

private void showDialogMessage(String msg) {
    // Simple AWT message dialog using a Frame-based Dialog
    Dialog dialog = new Dialog(this, "Notice", true);
    dialog.setLayout(new BorderLayout(10, 10));
    dialog.add(new Label(msg, Label.CENTER), BorderLayout.CENTER);
    Button ok = new Button("OK");
    ok.addActionListener(ev -> dialog.dispose());
    Panel p = new Panel();
    p.add(ok);
    dialog.add(p, BorderLayout.SOUTH);
    dialog.setSize(350, 130);
    dialog.setLocationRelativeTo(this);
    dialog.setVisible(true);
}

```

```
}

public static void main(String[] args) {
    EventQueue.invokeLater(FoodOrderingBillingSystem::new);
}
}
```

STEP 1: Customer / Order Details

Customer Name:

Table Number:

STEP 2: Food Selection

Food Category:

Food Item:

Quantity:

Add Item

Current Order:

Chicken Biryani x1 @ Rs.250.00 = Rs.250.00

STEP 3: Bill / Order Summary

```
=====
                        RESTAURANT FINAL BILL
=====
Customer Name : DEEPAK G
Table Number  : 3
-----
Ordered Items:
- Chicken Biryani x1 @ Rs.250.00 = Rs.250.00
-----
Subtotal      : Rs.250.00
GST (5%)      : Rs.12.50
-----
GRAND TOTAL   : Rs.262.50
=====
```